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Automatic Analysis of Basketball Shooting Based on Machine Learning

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***Abstract*—In order to help basketball players to carry out training more efficiently and reasonably, and obtain the technical specifications of their shootings timely in training, this paper constructs a shooting detection algorithm to assist basketball players in training. Firstly, Hough transform algorithm is applied to the shooting video to detect the trajectory of the basketball in the shooting. By analyzing the basketball trajectory, it is found that the best shooting angle should be 49° to 51°. Then, the basic principles of HOG feature and SVM are studied, and the HOG feature of basketball hit and miss is extracted to train the SVM classifier. Finally, the training model was tested, and the classification accuracy of basketball hit or miss was 91.75%. This study will help the basketball players in their training and also could be applied to other ball games.**

Keywords—Hough transform, HOG feature, SVM, motion analysis

I. Introduction

With the rapid development of computer vision technology, its application fields are constantly enriched, and it has penetrated into all aspects of people's lives. Sports analysis is also a bright spot in computer vision applications. In competitive sports, the standardization or error of the athletes' technical movements often directly affects the result of the competition. Therefore, athletes need to maintain a high level of skill and tactics through a lot of high-intensity training. In the training, through the camera device and video analysis technology, it can obtain the accurate data of the athletes' technical movements or the relevant target movement trajectory without additional interference, and put forward the scientific and reasonable guidance for the athletes' training after the analysis, so as to improve the training efficiency.

In the field of sports analysis, many research teams have already explored and studied relevant projects, such as the DVcoach system developed by the Australian team [1], which is used for the training of various ball players(such as strengthening the weak links of students' forehand and backhand movements in tennis classroom teaching [2]). Liu Hui [3] et al. analyzed the water polo shooting technique through video recording and revealed the basic rules and characteristics of its technical movements. We designed a set of basketball shooting training assistant algorithm, through shooting training videos of athletes and analysis of shooting track and hit situation, to help coaches to make more scientific and reasonable training guidance.

Hough transform detection circle method has strong anti-interference ability [4], and can still achieve ideal detection results even in the case of image blurring or partial region loss. Therefore, the Hough transform detection circle method is adopted in this paper to extract the discrete points on the basketball trajectory in the shooting video. The least-square method is used to fit the parabolic equation of shooting trajectory, and then the angle, velocity and the highest point of shooting trajectory can be calculated by analyzing the equation. And a large number of shooting videos were analyzed to get the range of beat shooting angle.

HOG feature has strong anti-interference capability [5] with adaptability to different light intensity. Support vector machine (SVM) has advantages of fast training speed, simple structure and global optimal, and the two are often used together in the field of target detection. We classified the basketball shot hitting based on the HOG feature of the basket and the support vector machine (SVM) algorithm, which is used to record the shooting hitting of athletes during training.

II. Shot trajectory detection

A. Circle detection

Hough transform is a kind of voting principle of parameter estimation technique, its principle is illustrated in references [6]. The region of image detection is transformed into parameter space in some way, and the process of finding circles is transformed into peak detection process by means of point voting and setting accumulator.

The equation of a circle is:

$$(x-a)^2+(y-b)^2=r^2 \tag{1}$$

In the formula (1), (a,b) is the center of the circle, and r is the radius of the circle.

Convert the circle on the $x-y$ plane of the image space to the $a-b-r$ parameter space, then a three-dimensional cone is formed in the parameter space corresponding to the circle passing through point (X,Y) in the image space, as shown in the figure 1 below:

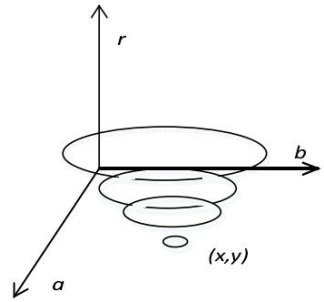

Figure.1 Parameter space

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The circle in the image space has a corresponding three-dimensional cone in the parameter space. Therefore, the circle with same point across the image space has a corresponding three-dimensional cone in the parameter space, and must intersect at a point (a, b, r) at the height of r.The parameters of the circle can be obtained from the detected point, so as to determine the circle.

B. Track recognition of basketball

The operation steps are as follows:

1) The video is decomposed into images frame by frame, and the edge of the input image is detected to obtain the boundary point, which is the foreground point.

2) The curve equation of the circle is transformed from $x - y$ coordinate system to $A - B$ coordinate system by coordinate transformation. I can write it as a minus x squared + b minus y squared =r squared

3) The points on the same circle in the original image must satisfy all the equations of the circle in the A-B coordinates after the conversion. It is intuitively shown that the circles corresponding to these many points will intersect at a point, then the intersection point may be the center of the circle(a, b).

4) Count the number of circles at the intersection point, and take the maximum value, then the coordinate of the center of the corresponding circle in the original image can be obtained.

Record the coordinates of the center of the circle $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ .Assume that the equation of parabola curve is $y = f(x)$, and construct the fitting curve is $p_k(x)$.

$$p_x(x) = a_0 + a_1 x + a_2 x^2 \tag{2}$$

Constitute loss function as:

$$e = \sum_{i=1}^{n} (y_i - a_0 - a_1 x - a_2 x^2) \tag{3}$$

The optimal undetermined coefficient satisfying the minimum loss function can be obtained by iterative gradient descent method. Assume$A = (a_0, a_1, a_2)$, and its initial value is (b_0, b_1, b_2).Then we can get :

$$\nabla e = \left(\frac{\partial e}{\partial a_0}, \frac{\partial e}{\partial a_1}, \frac{\partial e}{\partial a_2}\right) \tag{4}$$

Constitute iterative equation as:

$$A_{n+1} = A_n - \alpha \nabla e \tag{5}$$

α is step size. The iteration stops when $|A_{n+1} - A_n| < \varepsilon$ (ε is accuracy), and put A_{n+1} into $p_k(x)$ to get shot fitting parabolic trajectory.

III. SHOT HITTING DETECTION

A. HOG feature

HOG feature is based on the normalized local directional gradient histogram in dense mesh, which aims to describe the shape features of partial targets through the distribution of local gradient or edge direction. The eigenvectors formed by the combination of all HOG descriptors can be used to train the linear SVM and classify the samples[7]. HOG feature extraction is mainly divided into four steps below.

1) Gamma color normalization

Gamma normalization can reduce the local shadows and light changes of the image, thus effectively reducing the sensitivity of the algorithm to light. The Gamma compression formula is

$$I(x, y) = I(x, y)^{\gamma} \tag{6}$$

2) Gradient calculation

$G_x(x, y), G_y(x, y)$ represents the gradient operator in horizontal and vertical directions respectively, and the calculation formula is

$$G_x(x, y) = I(x + 1, y) - I(x - 1, y) \tag{7}$$

$$G_y(x, y) = I(x, y + 1) - I(x, y - 1) \tag{8}$$

$I(x, y)$is the gray value of the pixel, then the gradient amplitude and direction at the pixel can be expressed as

$$G(x, y) = \sqrt{G_x(x, y)^2 + G_y(x, y)^2} \tag{9}$$

$$\alpha(x, y) = \arctan\left(\frac{G_x(x,y)}{G_y(x,y)}\right) \tag{10}$$

3) Histogram of cell gradient direction

The image was segmented into several cell units, and the histogram of 9 bins was used to calculate the gradient information in the cell units. The gradient direction from 0° to 180° was divided into 9 direction blocks as shown in the Fig.2. The weighted projection of each pixel point in the cell unit is carried out in the histogram with the gradient direction. The weight is the amplitude of the pixel point gradient, and it is mapped into the corresponding Angle block to obtain the 9-latitude feature vector corresponding to the cell unit.

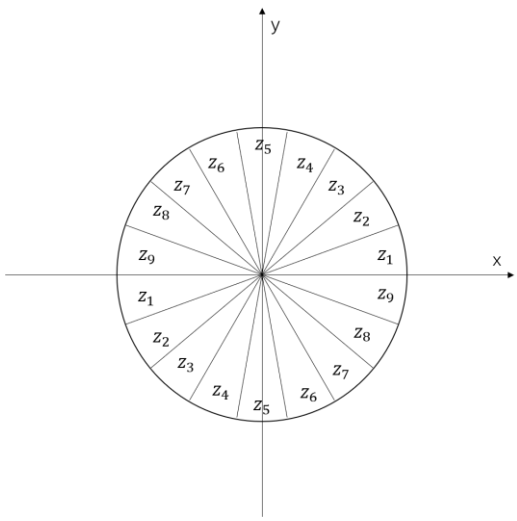

Figure.2 Diagram of gradient direction division

4) Intra-block gradient normalized histogram

In order to further compress light, shadow and edges features and make the eigenvector robust, cells need to be combined into spatially connected block. The eigenvectors of all cells in the block are connected in series and normalized to

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get the eigenvectors of the block. The division of block is shown in Fig.3. We adopt the $L1-norm$ normalized function

$$f=\frac{v}{\sqrt{||v||+\varepsilon}} \tag{11}$$

In the formula, f is the normalized eigenvector, v is the unnormalized block eigenvector, and ε is a minimal constant which aims to prevent the denominator from being zero.

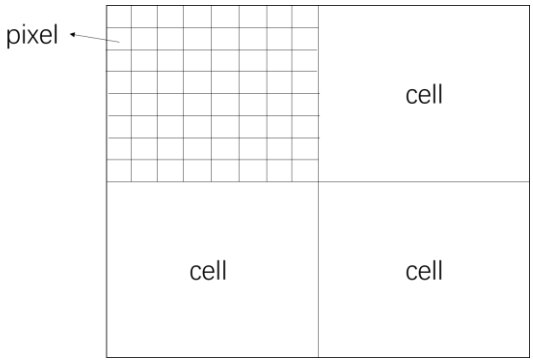

Figure.3 division of block

B. Support Vector Machine (SVM)

The core idea of SVC is to map the data set from the input space to the high-dimensional feature space through a nonlinear mapping function, and then solve the problem linearly in the feature space[8]. By constructing an optimal classification hyperplane, the distance between the nearest vector and the hyperplane is maximized, and the empirical risk and the confidence range of generalization are minimized, so as to realize the correct classification[9].

For a given training sample set $T=\{(x_i,y_i)\}, x_i\in R^n, y_i\in\{-1,+1\}, i=1,\dots,n$, look for a function $g(x)$ that divides a point on R^n into two parts. Fig.4 shows binary classification problems. l represents the classification line, lines l_1, l_2 represent the two lines closest to the optimal classification line, and w represents the distance between the two lines, namely, the optimal classification interval to be sought.

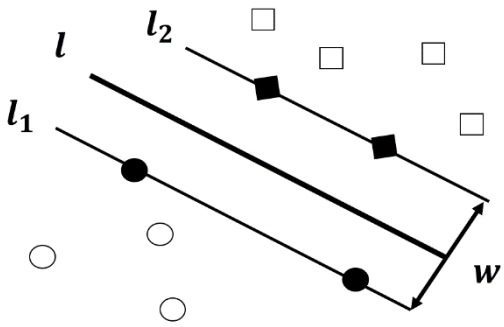

Figure.4 Schematic diagram of binary classifier

After normalization of the judgment function, the equations of lines l_1, l_2 can be expressed as $(w\cdot x_i)+b=\pm1$, where $(w\cdot x_i)$ represents the inner product of w and x_i.

The optimization problem can be obtained by using the maximum interval method:

$$\min_{w,b}\frac{1}{2}\|w\|^2$$

$$s.t.\, y_i\big((w\cdot x_i)+b\big)\geq 1, i=1,2,\dots,n \tag{12}$$

Assuming that the solution of the above equation isw^*, b^*, then $(w\cdot x_i)+b=0$ is the line between l_1 and l_2. In this case, the training sample points on the two lines l_1, l_2 are called support vectors[10].

In SVC, commonly used kernel functions include polynomial kernel function, radial basis kernel function, Sigmoid kernel function and linear kernel function. Different kernel functions can be used to construct different SVC[11]:

- Linear Kernel Function: $K(x,y)=x\cdot y$
- Polynomial Kernel Function: $K(x,y)=(x\cdot y+1)^d, d=1,2,\dots$
- Radial Basis Kernel Function: $K(x,y)=\exp(-\gamma\|x-y\|^2)$
- Sigmoid Kernel Function: $K(x,y)=\tanh(b(x\cdot y)-c)$

IV. Experimental and Results

A. Experimental environment

Hardware environment:Intel Core i5-8250U CPU @1.60GHz 1.8GHz, 8.0 GB running memory, NVIDIA GeForce MX150

Software environment:Python 3.7+ OpenCV 4.3.0.36

B. Shooting trajectory extraction and analysis

1) Trajectory extraction

The resolution of the detected video is 3840*2160, and the frame rate is 59.94 frames/second. The flowchart of trajectory extraction is shown in Fig.5.

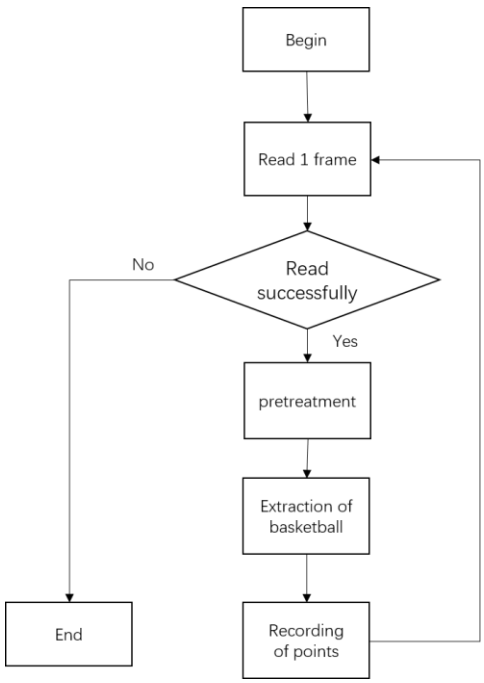

Figure.5 Flowchart of trajectory extraction

In preprocessing, the image is first transformed into a

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binary image, and then filtered to eliminate the influence of noise. Hough transform is used to detect the circular contour in the preprocessed image, and the threshold of the accumulator is set as 55, the minimum radius is 40, and the maximum radius is 70. A high accumulator threshold is set here to prevent certain areas of the background from being misidentified as circles. Since the actual motion track of the basketball is a continuous curve, the comparison link between the center position of the circle and that of the previous frame is added in the experiment, and the center of the circle that deviates too much or too little and the corresponding contour are discarded to eliminate the interference of the non-basketball circular contour. The trajectory extraction result is shown in Fig.6.

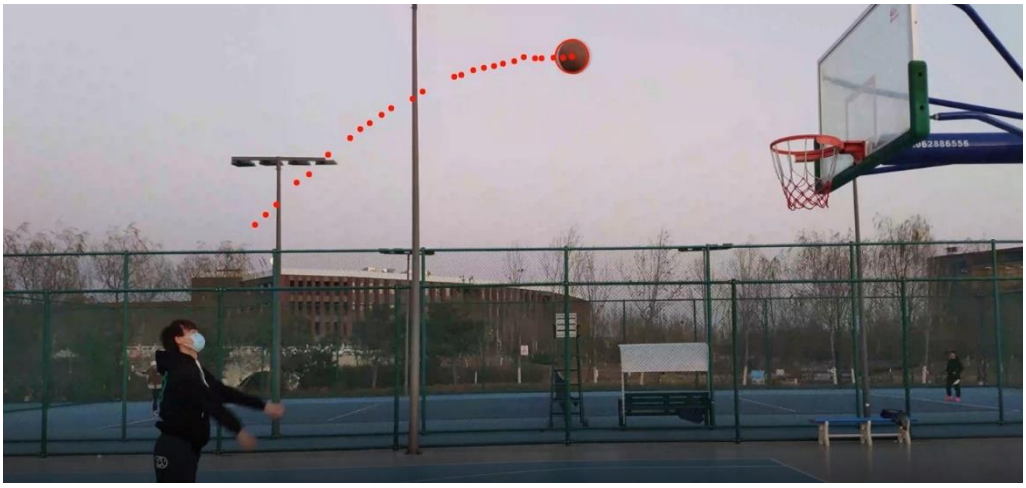

Figure.6 Trajectory extraction results

For the parabola $p_x(x) = a_0 + a_1 x + a_2 x^2$, the fitting results are as shown in table I.

TABLE I THE PARAMETER OF TRAJECTORY

a_2	a_1	a_0	angle/°
0.0013	-1.7836	624.5597	49.7847

2) Shot angle analysis

In this experiment, 63 shooting videos were analyzed, and shooting angles were obtained by fitting and merging the shooting trajectories. The shooting Angle distribution is shown in Fig.7.

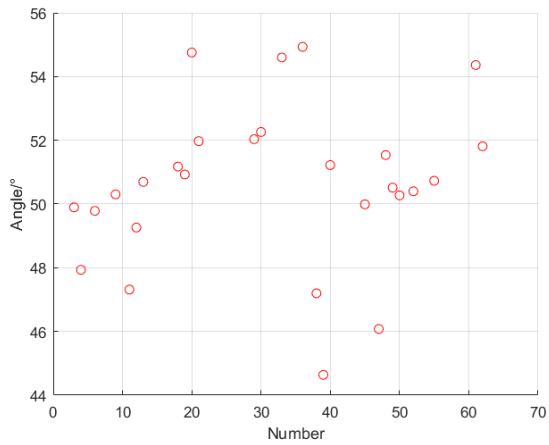

Figure.7 The angle distribution of the shot hitting

Make the shooting hit histogram and test the normality of shooting angle.

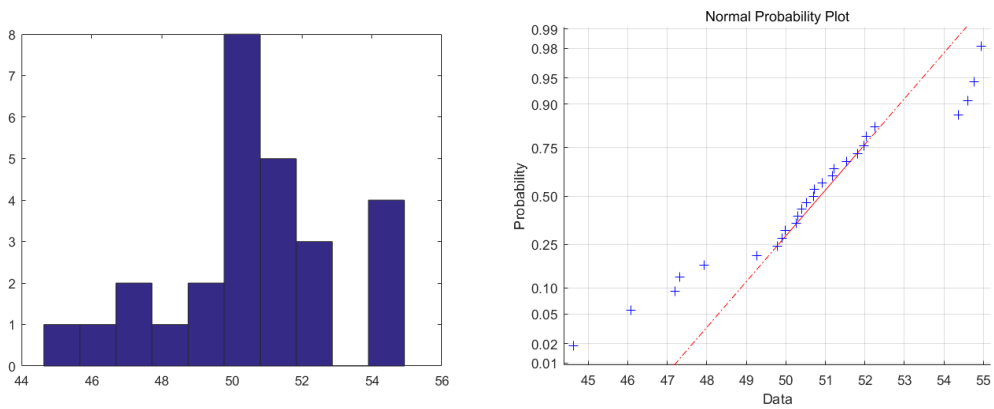

Figure.8 The results of the analysis

In the experiment, the shooting angle is roughly normally distributed, and the 95% confidence interval is [49.6177, 51.6113], which is shown in Fig.8.

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C. Shot hitting detection

The experimental procedure is as follows:

a) Making data sets and manually marking labels;

b) Extraction of HOG features;

c) HOG feature is applied to the training of SVM classifier;

d) Put the images of the detected area into the trained SVM model to get the predicted results and save them.

The flowchart of shot hitting detection is shown in Fig.9.

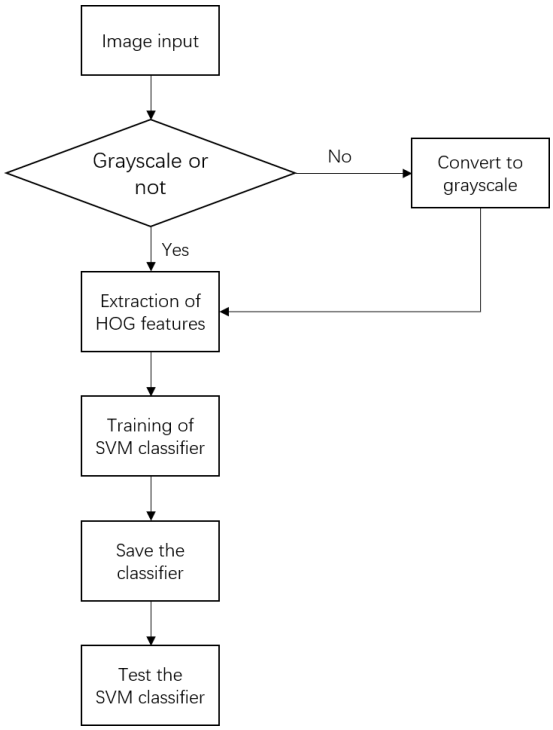

Figure.9 Flowchart of shot hitting detection

1) Data set making and HOG feature extraction

The data set in this experiment is manually annotated. Basket samples from different angles and positions are collected frame by frame using shooting videos collected on the Internet, including screenshots of scoring. The screenshots containing the goal pictures were marked as positive samples, with a total of 42. The rest were negative samples with a total of 55 pieces. The positive and negative samples are shown in Fig.10.

In the process of extracting HOG features of positive and negative samples used for training SVM model, cell contains 8*8 pixels, block are composite with 4*4 cells, and gradient direction angles are divided into 12. The extraction result is shown in Fig.11.

Figure.10 The positive and negative samples

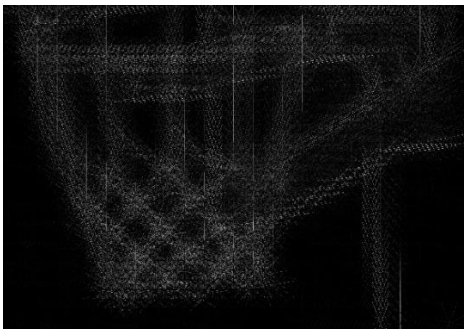

Figure.11 HOG feature extraction

2) Train the SVM classifier

The saved HOG feature is input as the training set, the linear kernel function is used to train the SVM classifier and the training results are saved. The training model was tested using a test containing 45 miss samples and 52 hit samples, and the classification accuracy was 91.75%.

V. Conclusion

We designed an algorithm to analyze the shooting of basketball players, which can be used to assist training. The experimental results show that the algorithm can identify the

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shooting trajectory accurately, and the classification accuracy of the hitting can reach 91.75%. And through the data analysis of shooting angle, we found that the best shooting angle should be between 49° and 51°. It can help athletes adjust shooting posture and improve training efficiency by analyzing the shooting trajectory of athletes in training and combining with the best angle. In addition, the algorithm can be deployed on the embedded platform to connect with the camera device, or directly connect the camera device with the superior machine so as to realize real-time analysis and supervision of athletes' training.

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